

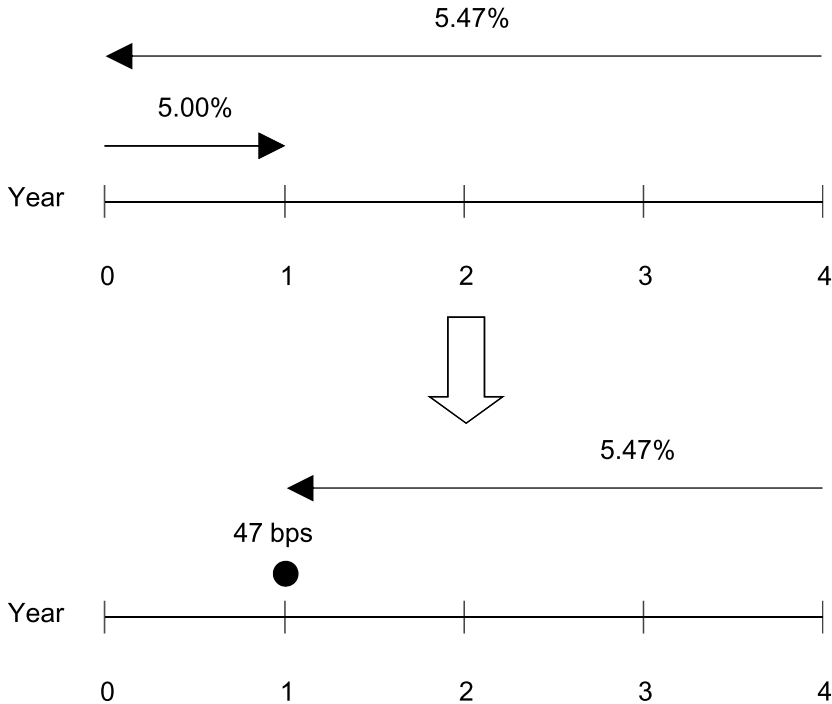


Figure 3.16: The Net Carry from Receiving in Four Years and Paying in One Year

and will thereafter receive the rate of 5.47% at the end of years two, three, and four.³⁷ This is demonstrated in Figure 3.16. So this replicating strategy almost gets us cash flows in all the right places (meaning at the same times) as the forward starting swap. The only problem is that this strategy earns *carry* (cash flow) over the *delay* (the period of time between the normal $T+2$ settlement date of a spot starting swap and the effective date of the forward starting swap). In this case, the replicating strategy earns carry of 47 bps over the delay of year one. The “•” in Figure 3.16 represents the carry that is earned in year one before the forward starting swap is effective.

As suggested above, we can solve for an annuity to be paid at the end of years two, three, and four that is worth the same in present value terms as

³⁷We are talking here about receiving rates and bps, but we shouldn't forget that there is some underlying notional upon which these are all being paid. It may be more comforting to think of this in terms of actual cash flow. In that case, pick a representative notional, say, \$100mm, and translate the 47 bps to \$470,000 and the 5.47% to \$5,470,000.

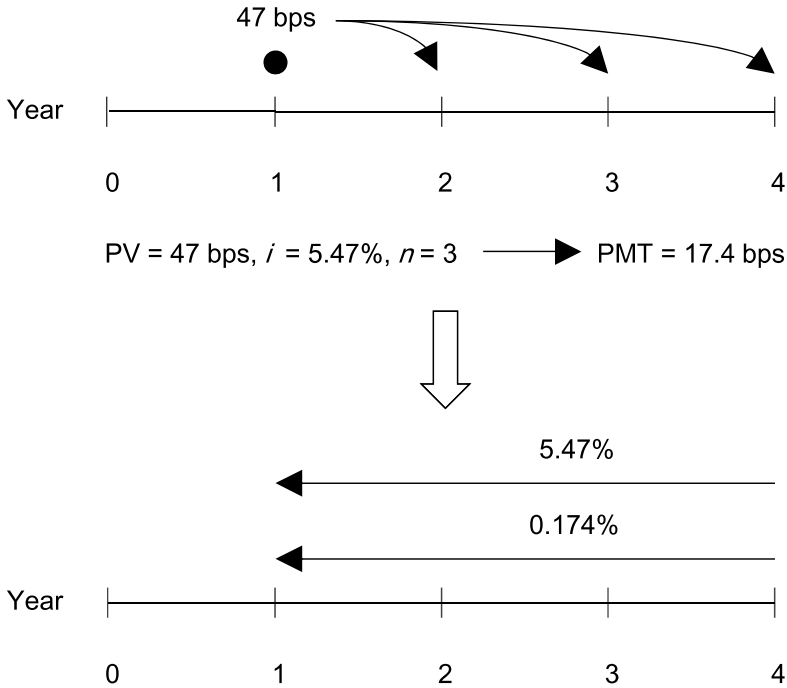


Figure 3.17: Moving the Net Carry from the Effective Date over the Life of the Swap

the 47 bps received at the end of year one. We need to do this because if this strategy is to replicate the forward starting swap, then it must not have any cash flows in the first year. This procedure is demonstrated in Figure 3.17. So instead of regarding this replicating strategy as producing carry of 47 bps at the end of year one, we can instead regard it as producing an annuity of 17.4 bps over the back three years. But this means that our handcheck for the 1x4 swap rate is $5.47\% + 0.174\% = 5.644\%$.

Why? Let's review what we just did. First, remember that the 1x4 swap rate is that fixed rate that—when paid over three years, one year forward—has the same present value as the LIBOR cash flows paid during the same time period. Now, if we enter into the replicating strategy, we will replicate the LIBOR cash flows in the forward starting swap, since the first year LIBOR cash flows cancel out, and we are left with the three LIBOR cash flows, one year forward. So we're halfway there. Now, on the fixed side, we need to come up with a single fixed rate over the forward three years. By entering into the replicating strategy, we are left with 47 bps

received at the end of one year, plus we get a rate of 5.47% received over the back three years. But as we saw, we are indifferent between receiving that 47 bps at the end of year one or receiving an annuity of 17.4 bps over the back three years. Thus, by receiving in four years and paying in one year, we have an annuity of 17.4 bps over the back three years plus a rate of 5.47% over the back three years, for a total of $5.47\% + 0.174\% = 5.644\%$ over the back three years. Note that this is the answer we got from our model, 5.644%, that we priced in Figure 3.14.

There is one thing that we forgot to mention. In Figure 3.17, when we took the 47 bps and solve for the annuity, we used a rate of 5.47% to discount. Why did we use 5.47%? We have no good reason, other than it was a readily available rate to use. We know by definition that any rate we use is wrong, since we should be using zero coupon rates to discount individual cash flows. But this is a handcheck, and we want a quick and dirty approximation, so we live with it. We need a rate to discount cash flows received over three years (actually three years, one year forward), so it seemed reasonable to use the 4-year swap rate. Is that “right”? No, of course it isn’t right. But this is just a handcheck—we’re not going to trade off of this number; we are just going to use it to make sure we are in the neighborhood of the right answer. $\square$

Example 2

Let’s take a look at another example that is slightly more complicated but makes use of the same exact methodology that we just employed to calculate the 1x4 forward starting swap rate. Now let’s consider a 2x5 (i.e., a 3-year swap, two years forward), and assume that the fixed side pays semi bond and the floating side is 3-month LIBOR quarterly money. Suppose that the swap curve is as given in Table 1.1. Note that the mid-market 2-year swap rate is 0.685%, and the mid-market 5-year swap rate is 1.627%.

This example won’t fit into our model that has annual cash flows, so we will resort to a handcheck in order to compute the 2x5 swap rate. A dealer who receives fixed in a 2x5 swap can hedge the position by paying fixed in five years and receiving fixed in two years. A replicating strategy of receiving the 2x5 swap rate entails receiving in five years and paying in two years. Figure 3.18 shows this replicating strategy.

The important thing to keep in mind is that the replicating strategy will imply what the fixed rate in the 2x5 swap must be. In the replicating strategy, the two LIBOR legs will exactly cancel each other out over the first two years (i.e., the first eight 3-month LIBOR flows that are paid out in the 5-year swap will exactly cancel out the eight 3-month LIBOR flows that are received in the 2-year swap), and the dealer will be left paying LIBOR only over the back three years. This is what we want: in the forward starting

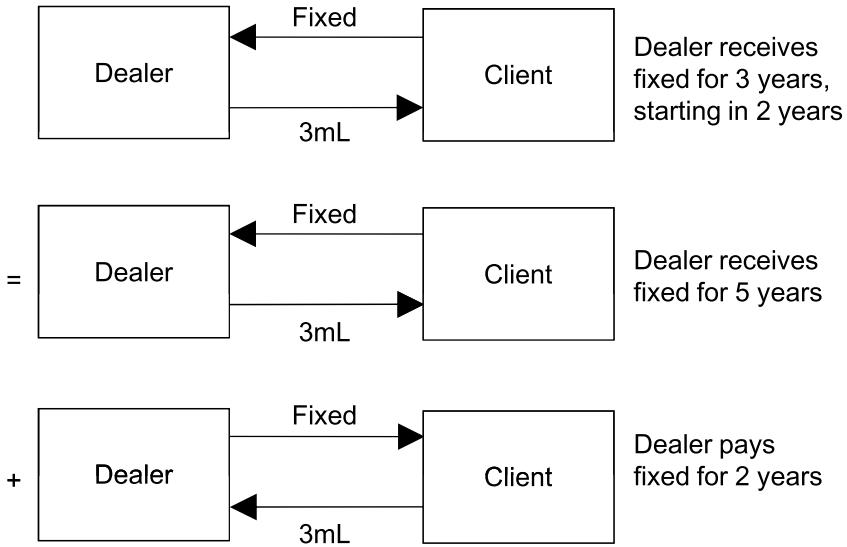


Figure 3.18: Replicating a 2x5 Swap

swap the dealer also pays LIBOR for three years with an effective date two years from now. So by entering into the replicating strategy, we emulate the floating leg of the swap perfectly.

Let's now look at the fixed flows that result from the replicating strategy. Since the two fixed rates differ, this means that the dealer earns carry of 94.2 bps (the difference in the two fixed rates of 1.627% and 0.685%) running over the first two years and will receive the rate of 1.627% for the next three years. This is demonstrated in Figure 3.19. So this replicating strategy nearly emulates the cash flows that occur in the forward starting swap. The only problem is the *carry* over the delay of the first two years, which is $1.627\% - 0.685\% = 94.2$ bps. Let's be very, very careful: this carry is paid out semiannually at $94.2/2 = 47.1$ bps per semiannual period, and it is very important that we value it as such.

We need to take all this carry over the delay (i.e., the period of time between when the swap is transacted and when it is effective) and shove it into the back three years. We will do this in two steps. In step one, we are going to take the four annuity payments in the first two years and compute their future value (FV) as of the end of year two, which is the effective date of the forward starting swap. In step two, we are going to take the value of everything we just pushed forward to the effective date and spread it out as an annuity over the next three years, which is when the forward starting swap is in existence. These procedures are shown in Figure 3.20. So instead

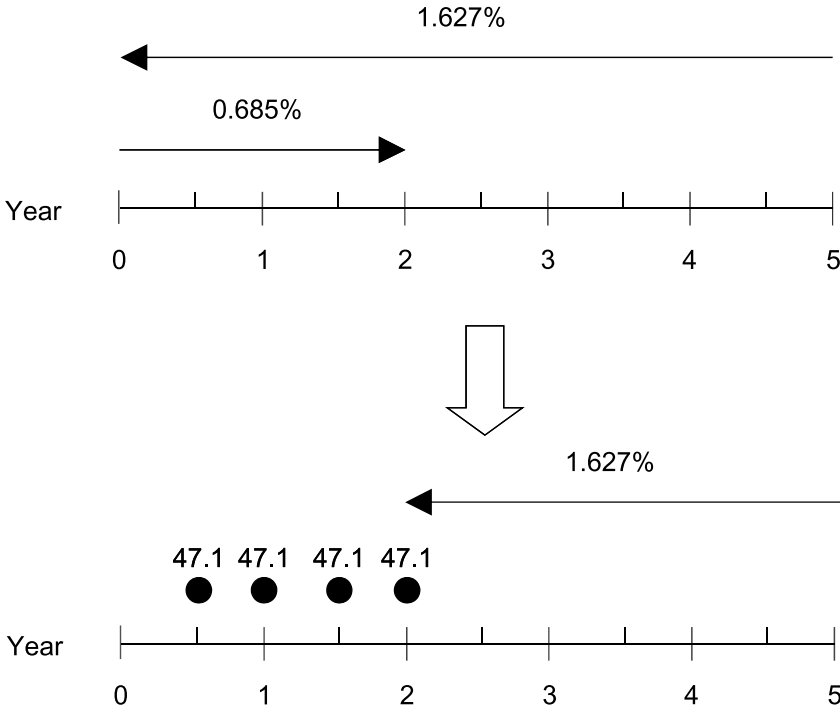


Figure 3.19: The Carry from Receiving in Five Years and Paying in Two Years

of regarding the replicating strategy as producing carry of 94.2 bps running over the delay, we can instead regard it as producing an annuity of 32.5 bps in semiannual installments over the back three years. But this means that our handcheck for the 2x5 swap rate is $1.627\% + (2 \times 0.325\%) = 2.277\%$. This is shown in Figure 3.21. Let's be very, very careful to note that when we take the annuity of 32.5 bps per semiannual period, we must multiply it by two in order to bring it back to annualized terms before we add it to 1.627%.³⁸ □

³⁸Note that in Figure 3.20, we use the 2-year swap rate of 0.685% as a discount rate for purposes of computing the future value of the carry over the delay, and we use the 5-year swap rate of 1.627% as a discount rate for purposes of translating this future value into an annuity over the life of the forward starting swap. In general, we would use the shorter-tenor swap rate for the purposes of computing the future value of the carry over the delay and the longer-tenor swap rate for the purposes of translating this future value into an annuity over the life of the forward starting swap.

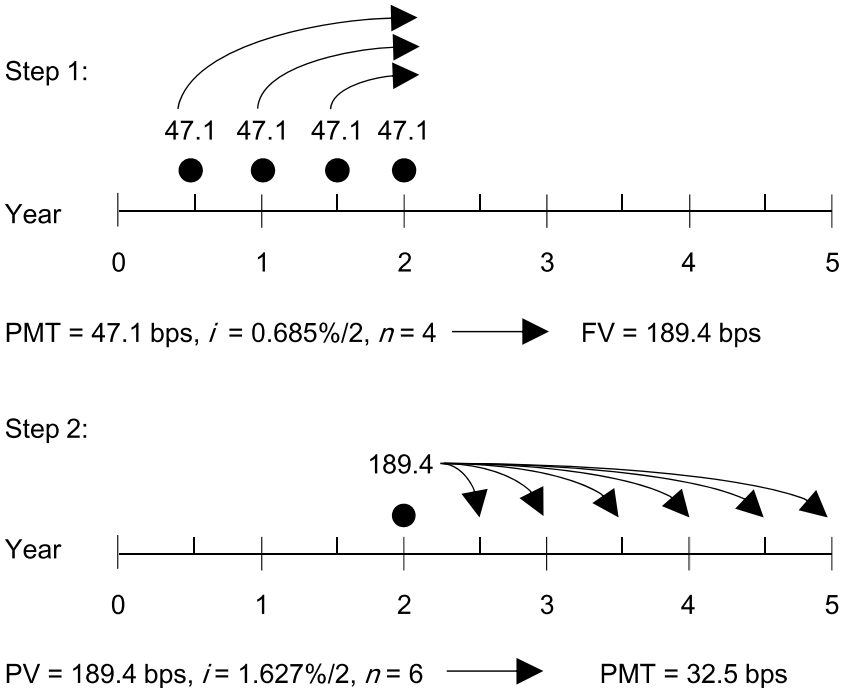


Figure 3.20: Taking the Future Value of the Carry Over the Delay and Spreading It Out Over the Life of the Swap

To summarize, here is the suggested procedure to approximate the forward swap rate from the two relevant spot starting swap rates:

1. Calculate the carry from the two spot starting swaps between today and the effective date of the forward starting swap (a period of time often referred to as the *delay*).
2. Compute the forward value of this carry to the effective date of the swap, using the swap rate on the shorter-tenor swap as discount rate.
3. Take this amount and calculate an annuity over the life of the forward starting swap, using the swap rate on the longer-tenor swap as discount rate.
4. Add the annuity (annualized in the case of multiple swap payments per year) to the longer-tenor swap rate.

Step 2:

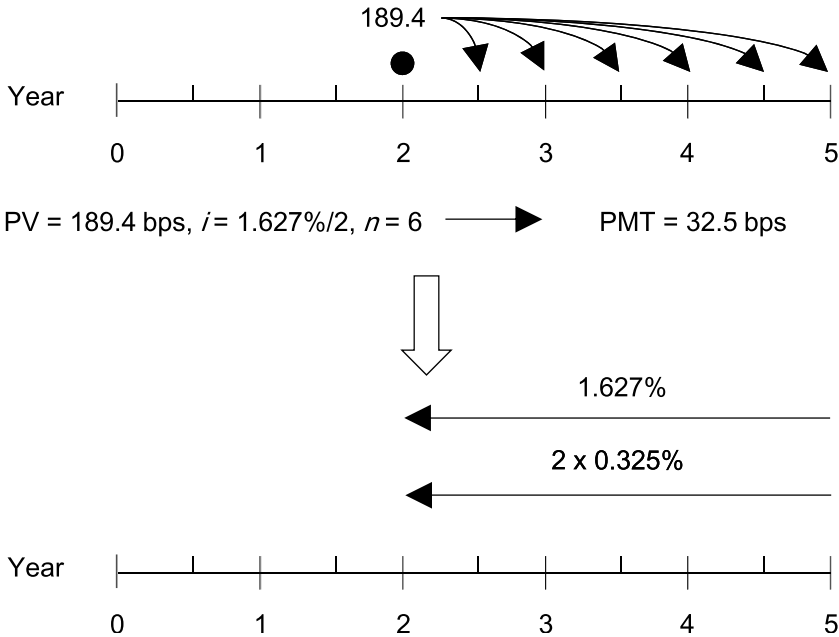


Figure 3.21: Obtaining the 2x5 Swap Rate from the Annuity Computation

Uses of Forward Starting Swaps

There are a number of ways in which investors use forward starting swaps. We will first look at a classic example in which corporations use forward starting swaps (i.e., by swapping an anticipated asset or liability that is expected to commence some time in the future). We will also discuss how forward starting swaps have become market benchmarks in more recent times.

Hedging Future Issuance

The most traditional use of a forward starting swap is to hedge an asset or liability that does not commence today but is anticipated to start at a known point of time in the future. Consider, for example, a corporation that knows it is going to need to raise funds in a year to undertake an investment project. The corporation is going to want to raise \$500mm by issuing a 5-year fixed rate bond. The corporation does not want to issue the bond now, because it does not need the cash now.

When the corporation comes to market in a year to issue the bond, the yield that it will pay to the bondholders will be a function of both 5-year on-the-run Treasury yields prevailing at the time, T_5 , as well as the corporation's corporate spread, c_5 , prevailing at the time. That is, the corporation will pay a yield of

$$T_5 + c_5. \quad (3.9)$$

If the sum in Equation 3.9 increases over the next year, this will serve to increase the funding costs of the corporation. Of course, this sum may decline next year as well. But the point is that the corporation is exposed to movements in this sum; the yield on this bond will depend on the sum in Equation 3.9 in a year.

The corporation may be concerned that their funding costs are going to increase when it is time to come to market in a year. It may choose to execute a swap to hedge against this risk. What's the best swap hedge for the corporation? If it enters into a 5-year spot starting swap, that will produce cash flows over the next year, whereas its anticipated debt does not. Moreover, this hedge will not provide coverage for the last year that the debt is outstanding. If it enters into a 6-year spot starting swap, although this hedge would cover the entire life of the anticipated liability, we would again have the problem of the hedge existing in the first year, whereas the debt does not.

It turns out that a better hedge would be for the corporation to enter into a forward starting swap. In particular, the corporation might enter into a 1x6. This is a 5-year swap, one year forward, which matches the timing of the anticipated debt. The corporation will want to pay fixed in the 1x6. The corporation will hold this swap for one year, at which time it will unwind the swap. Let's assume that the fixed rate in the 1x6 that the corporation pays is 5%. There will be no cash flows over the course of the first year. Note that in one year's time, the swap will then be a 5-year spot starting swap. If, in one year, 5-year swap rates are above 5%, the swap will be in-the-money to the corporation. The profits on the swap can be used to offset the (presumably) higher funding costs associated with the debt. And if 5-year swap rates are below 5% in one year, the loss on the swap will (presumably) be met with lower funding costs on the debt. Again, the corporation will unwind the swap in one year, recognizing the profit or loss on the swap at the end of the hedging period, and use this gain or loss to offset its funding costs.

Notice that we used the word "presumably" twice in the preceding paragraph. This is because the swap being in-the-money does not guarantee that the funding costs of the corporation will be higher in a year than they would be today, and the swap being out-of-the-money does not guarantee that the

funding costs of the corporation will be lower in a year. In one year, the value of the swap that the corporation entered into will depend entirely on the then-prevailing 5-year swap rate. We do not know where the 5-year swap rate will be in one year. Its value is random and can be expressed as

$$T_5 + ss_5, \quad (3.10)$$

where ss_5 represents the unknown 5-year swap spread prevailing at the time. If we compare the random funding costs of the corporation in Equation 3.9 with the random 5-year swap rate in a year as given in Equation 3.10, the former depends on the corporation's random funding spread, the latter on 5-year swap spreads. Presuming the corporation's 5-year corporate spread is positively correlated with swap spreads, the hedge will likely be more effective than if there is negative correlation between the two. This dynamic is demonstrated in Figure 3.22.

There may be good reason to suggest that the corporation's credit spread is positively correlated with swap spreads. As we discussed in Chapter 2, swap spreads tend to be a barometer of general credit concerns in the market. Therefore, an increase (decrease) in the corporation's credit spread due to a systemic change in credit spreads may be accompanied by an increase (decrease) in swap spreads. The corporation may wish to analyze the historical correlation of its credit spread to swap spreads in determining whether or not the swap is a good hedge.

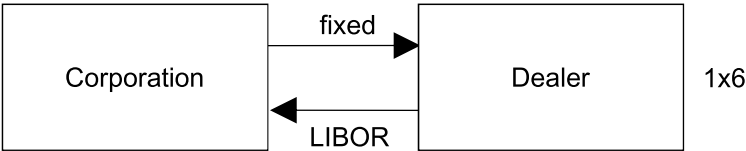
Forward Starting Swaps as a Benchmark

Most every market has its benchmarks, or barometers, that people watch to gauge the market overall. In the Treasury market, for example, the yields of the on-the-run 2-year, 10-year, and 30-year Treasuries have all been popular benchmarks. The interest rate swaps market has its benchmarks as well. The 2-year, 10-year, and 30-year swap rates are each watched by many investors to give a quick snapshot of the market. In recent years, certain forward starting swaps have become benchmarks as well.

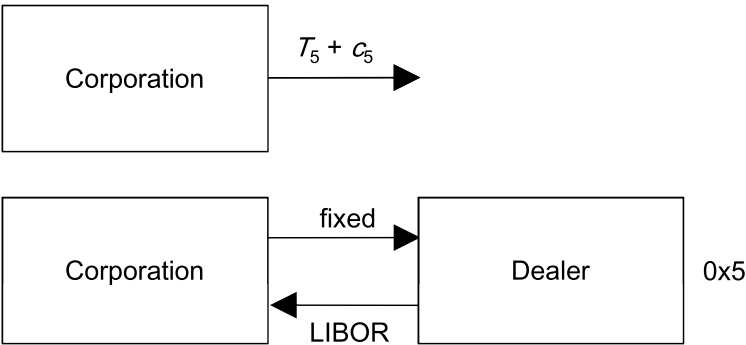
If we think about a 10-year spot starting swap (with standard conventions of semi bond versus threes) and how it is priced, it is composed of forty 3-month LIBOR forwards, each of which contributes to the value of the 10-year swap. The nearer-term forwards are very much tied to expectations of Fed policy, whereas longer-term forwards are less tied to the Fed's policy decisions and are more tied to the longer-run prospects for inflation, the economy, and so on.

In addition to looking at 10-year spot starting swaps, some investors in the market have become just as focused on looking at the 5x10 swap rate

Step 1: In anticipation of issuing a 5-year fixed rate bond in a year, corporation pays fixed on a 1x6 today:



Step 2: In a year, corporation issues fixed rate bonds at then prevailing market rates, $T_5 + c_5$, and unwinds swap:



Rates in <u>1 year</u>	Fixed Coupon <u>on Bond</u>	Value of <u>Swap Hedge</u>
High	High	Positive to Corporation
Low	Low	Negative to Corporation

Figure 3.22: Corporation Pays Fixed in a 1x6 to Hedge Future Issuance